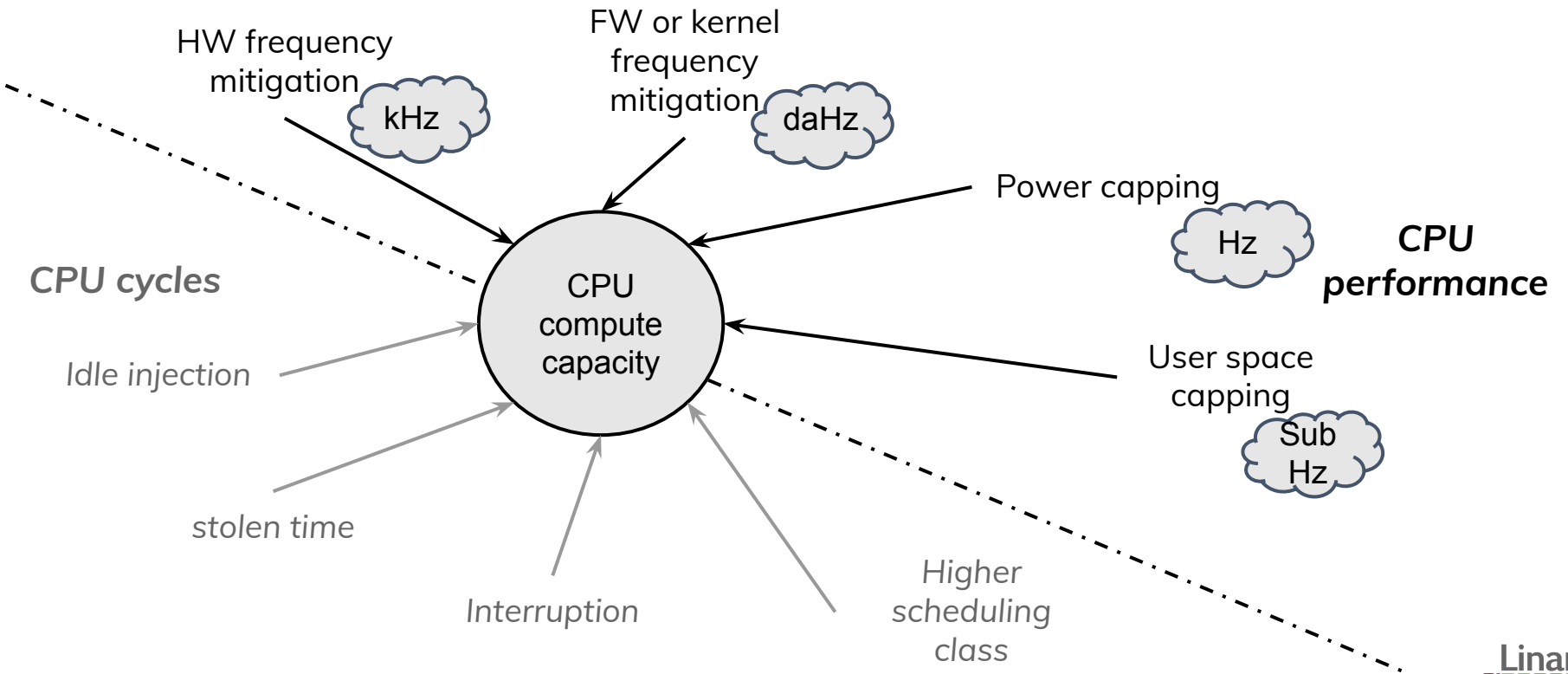
System pressure and CPU capacity

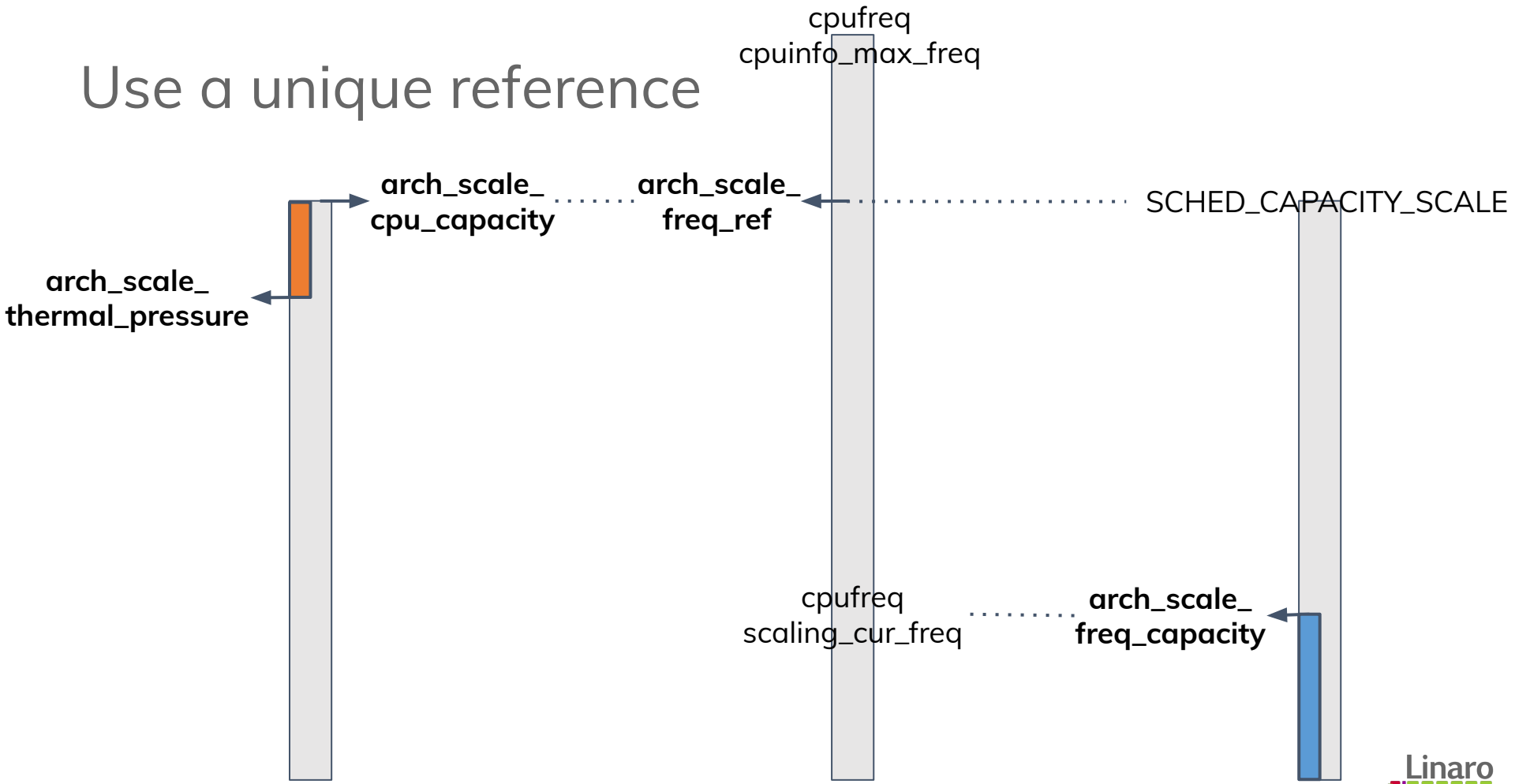
Vincent Guittot
Linaro Ltd
Nov 12th 2023



What does impact the compute capacity of CPU?



Use a unique reference



Linaro

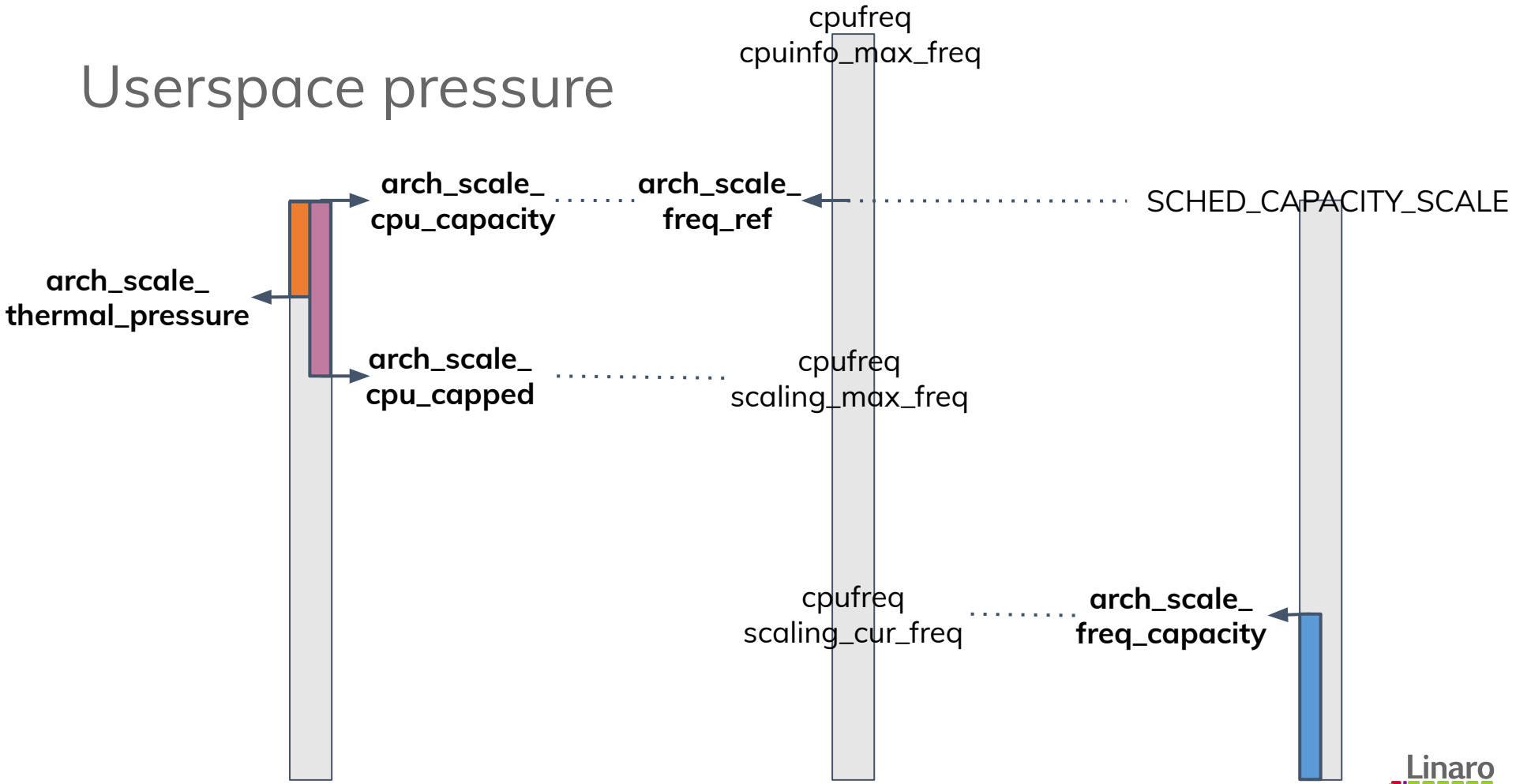
Userspace pressure on CPU capacity

- /sys/../cpufreq/policy*/scaling_max_freq
  - Store: Set **max_freq_request** freq_qos
  - Show: Return min of all FREQ_QOS_MAX requests

- Feedback the userspace max_freq_request into the scheduler
  - Can’t reuse arch_scale_cpu_capacity
  - Add a new arch_scale_cpu_capped

- One concern could be misfit_task detection
  - Rebuild sched domain but expensive and not scalable

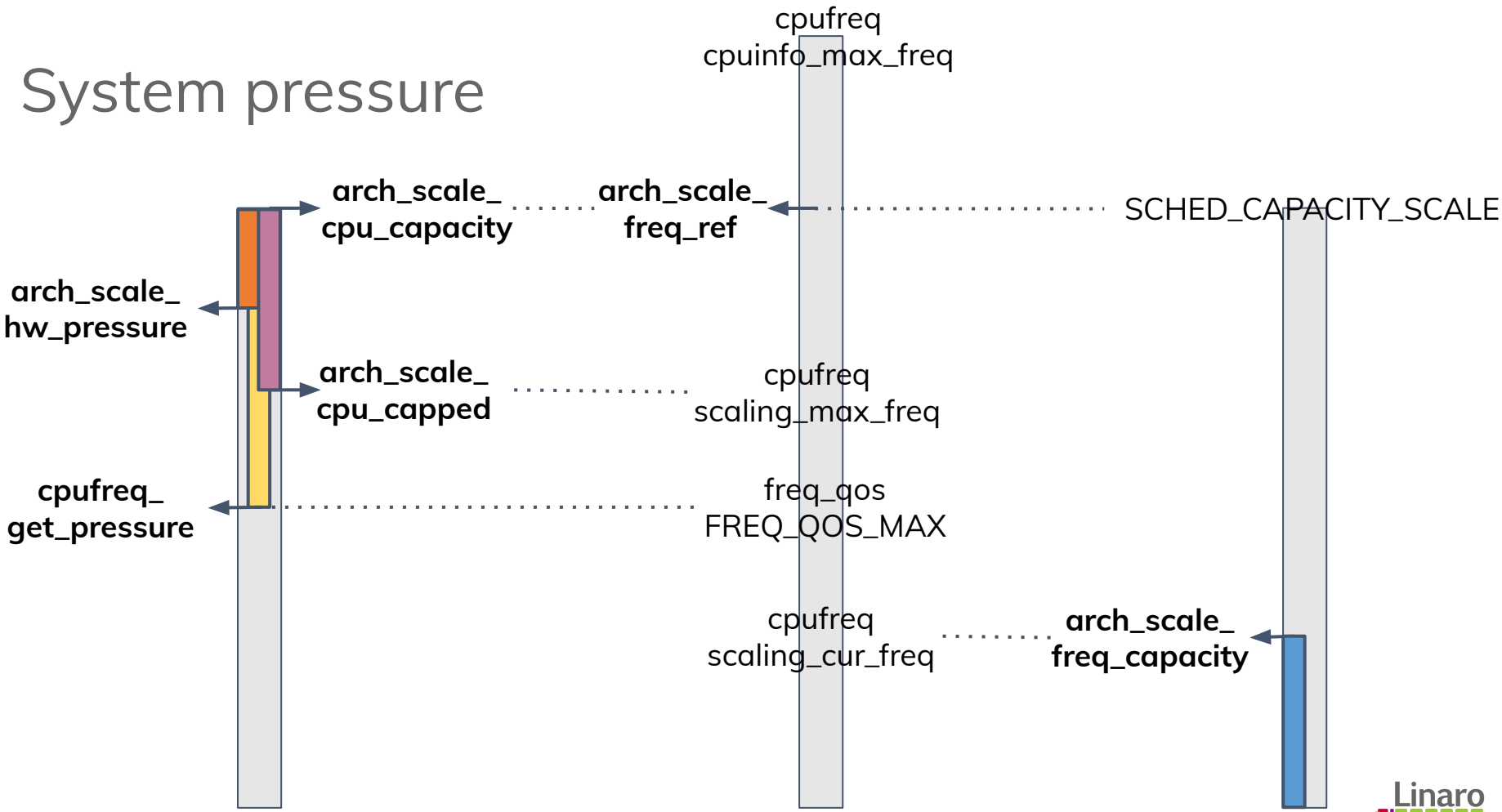
Userspace pressure



Firmware/kernel pressure on CPU capacity

- Need to handle several requests
  - And get the maximum pressure applied on the CPU
- Use FREQ_QOS_MAX requests ?
  - Blocking notifiers can be a problem
  - Schedule a work to handle the update by cpufreq policy
- Keep arch_scale_thermal_pressure for HW only
  - Rename it arch_scale_hw_pressure

System pressure



Get global system pressure on CPU capacity

- Get the lowest capacity among
  - arch_scale_cpu_capped
  - FREQ_QOS_MAX requests feedback
  - thermal_load_avg (to be renamed)

Thank you

